ICT2142 E- Business Systems

Lecture 04

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Designing & Developing E-Commerce Websites



Learning Outcomes



01 **Core Functions of an E-Commerce System**

The essential front-end and back-end building blocks every online store needs.



02 **UX/UI Design Principles**

Design principles that build trust and guide customers smoothly toward purchase.



03 **Information Architecture & Navigation**

Structuring content and menus so products are easy to find.



04 **Product Catalog, Search & Filtering**

Designing listings, detail pages, search and faceted filtering.



05 **Responsive & Mobile-First Design**

Adapting layouts across devices, with mobile as the starting point.

What Is an E-Commerce Website?

A digital platform that lets a business showcase, sell and manage products or services online — covering the full customer journey from product discovery to payment and delivery.

Combines a customer-facing storefront with back-office systems for inventory, orders and payments.

Must work reliably for many simultaneous users, across many devices and locations.

Success depends as much on design and usability as on the underlying technology.



Always On

The storefront operates continuously, worldwide, with no fixed opening hours.



Data-Driven

Clicks, searches and purchases can be tracked to continuously improve the experience.



Multi-Device

Customers move between desktop, tablet and phone within a single shopping journey.



Core Functions of an E-Commerce System

Front-End: Customer-Facing Functions

The features a shopper interacts with directly while browsing and buying.



Product Browsing & Catalog

Category pages and listings that display products with images, names and prices.



Search & Filtering

Tools that help shoppers locate specific products quickly within a large catalog.



Shopping Cart

A temporary holding area for items a customer intends to purchase.



Checkout & Payment

The guided process that turns a cart into a confirmed, paid order.



User Accounts

Registration, login, saved addresses, order history and wishlists.



Customer Support

Live chat, FAQs and contact options that resolve doubts before they cause drop-off.

Back-End: Administrative & System Functions

The behind-the-scenes systems that keep the storefront running and up to date.



Inventory Management

Tracking stock levels in real time so unavailable items aren't sold.



Order Management

Processing, packing, shipping and handling returns or exchanges.



Content Management (CMS)

Updating product details, pricing, banners and promotions without new code.



Payment Gateway Integration

Securely connecting to processors such as PayHere, Stripe or PayPal.



User & Access Management

Controlling admin roles, permissions and customer account data.



Analytics & Reporting

Measuring traffic, sales trends and conversion rates to guide decisions.

Payment Processing & Security



Every completed sale depends on this hand-off being both smooth for the customer and secure end-to-end.

SSL / TLS Encryption
Encrypts data in transit between the customer's browser and the server.

PCI-DSS Compliance
Industry standard for handling and storing cardholder data safely.

Tokenization
Replaces sensitive card details with a non-reversible token for storage.

Fraud Detection
Flags unusual transaction patterns before an order is confirmed.

UX/UI Design Principles for E-Commerce

UX vs UI in E-Commerce



UX (User Experience)

How easily and confidently a customer can complete a task.

Example: finding a laptop, comparing options and checking out without confusion.



UI (User Interface)

The visual and interactive layer.

Example: buttons, typography, colors, product cards, icons, spacing and form controls.

Core Design Principles

Simplicity & Clarity
Minimise cognitive load — clear labels, short paths, no unnecessary steps.

Consistency
Uniform colours, fonts and button styles across every page and screen.

Visual Hierarchy
Guide the eye to what matters most: product, price, and call-to-action.

Immediate Feedback
Loading states, confirmations and clear error messages at every action.

Accessibility
Sufficient colour contrast, alt text and full keyboard navigation.

Trust & Credibility
A professional, polished look signals a safe place to enter payment details.

Visual Design Elements

Colour

Use brand colour purposefully; reserve one strong accent colour exclusively for calls-to-action like “Add to Cart”.

Typography

Limit pages to two font families; keep prices and headings legible at a glance.

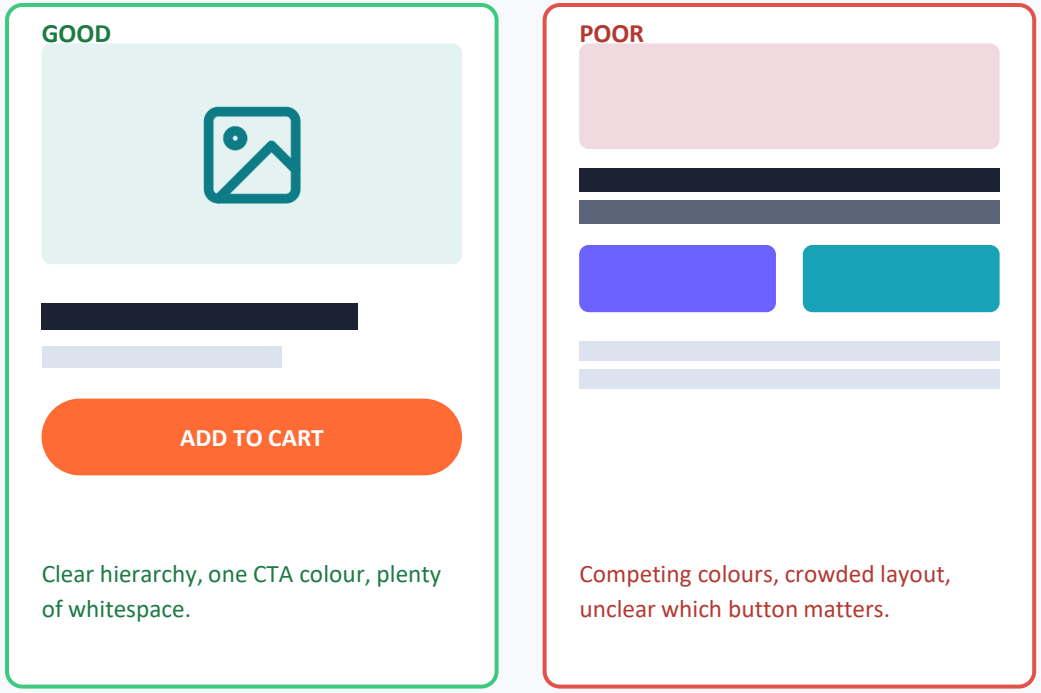
Whitespace

Avoid clutter — generous spacing lets product images and key details breathe.

Imagery

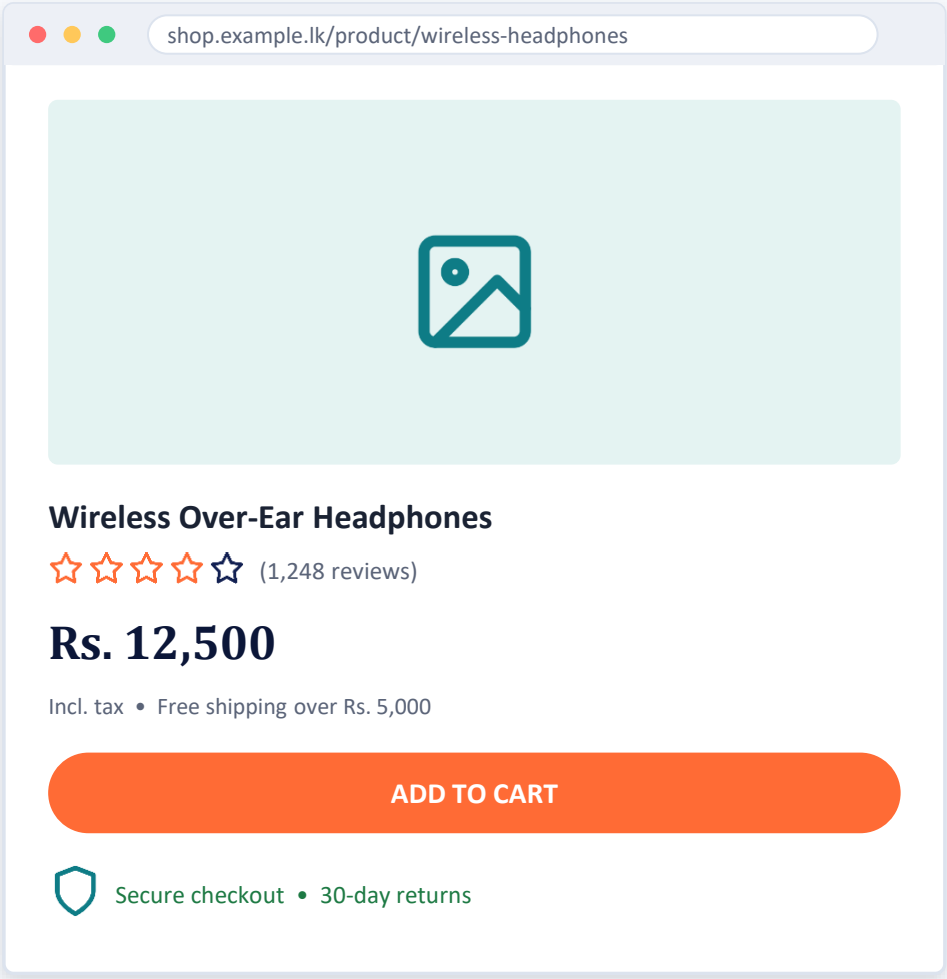
High-quality, consistent product photography raises perceived value and trust.

A QUICK COMPARISON



Building Trust & Reducing Friction

- Trust badges & secure checkout icons visible near payment fields.
- Genuine customer reviews and star ratings on product pages.
- Clear, prominent calls-to-action — one primary action per screen.
- Transparent pricing — shipping and tax shown early, not at the last step.
- Guest checkout available — registration is optional, not forced.
- Return and refund policy is easy to find before purchase.





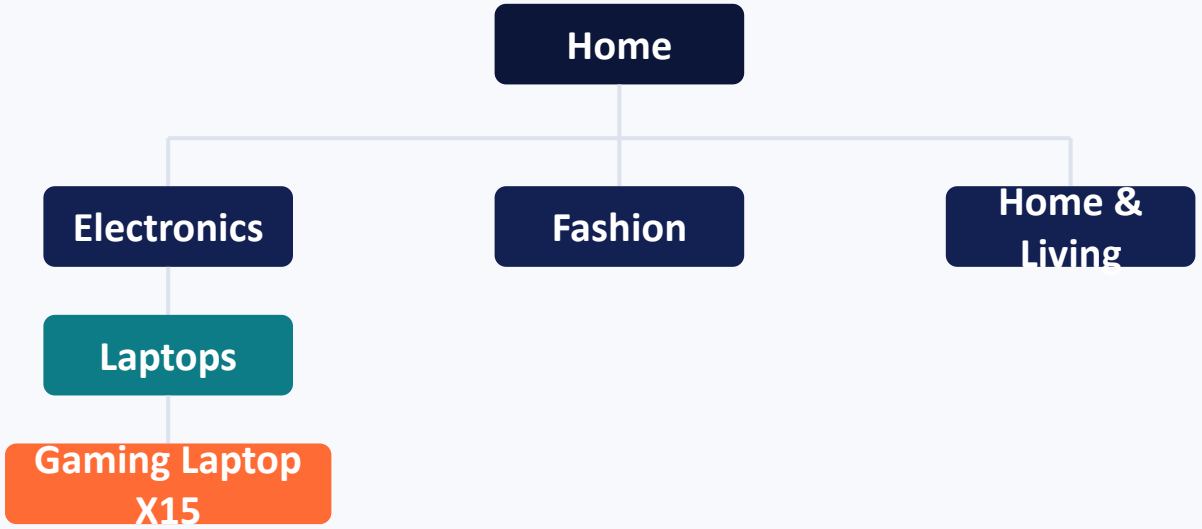
Information Architecture & Navigation Design

Structuring Content

Information architecture (IA) is the practice of organising, structuring and labelling content so it is findable and usable.

- Logical categorisation that matches how customers think about products.
- Shallow hierarchy — keep any product reachable within 2-3 clicks.
- Consistent labelling of categories, filters and navigation terms.
- Scalable structure that accommodates new products and categories.

EXAMPLE SITE HIERARCHY



Home › Electronics › Laptops › Gaming Laptop X15

Navigation Design Patterns

Global Navigation

Primary categories stay visible on every page.

Mega Menus

Multi-column dropdowns organise a large catalogue at a glance.

Breadcrumbs

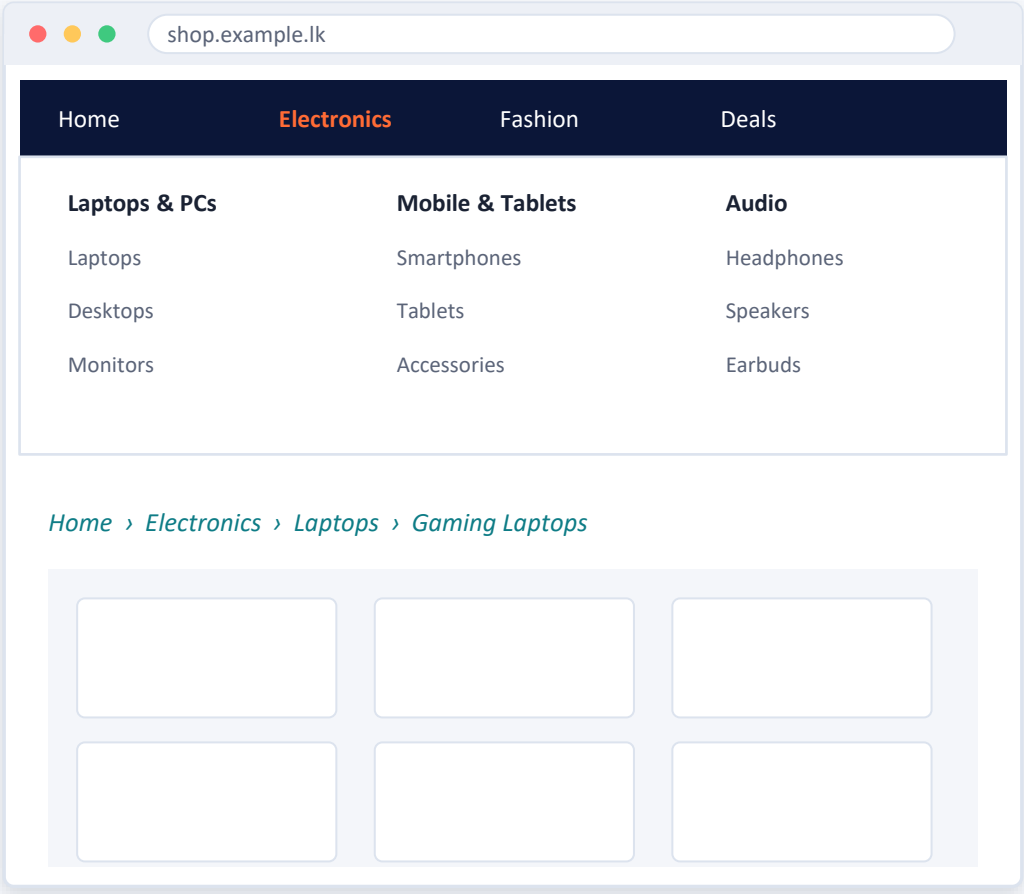
Shows the current location and lets users step back easily.

Sticky Navigation

Menu and cart icon stay visible while the page is scrolled.

Footer Navigation

Secondary links — policies, contact, sitemap — live in the footer.





Product Catalog, Search & Filtering

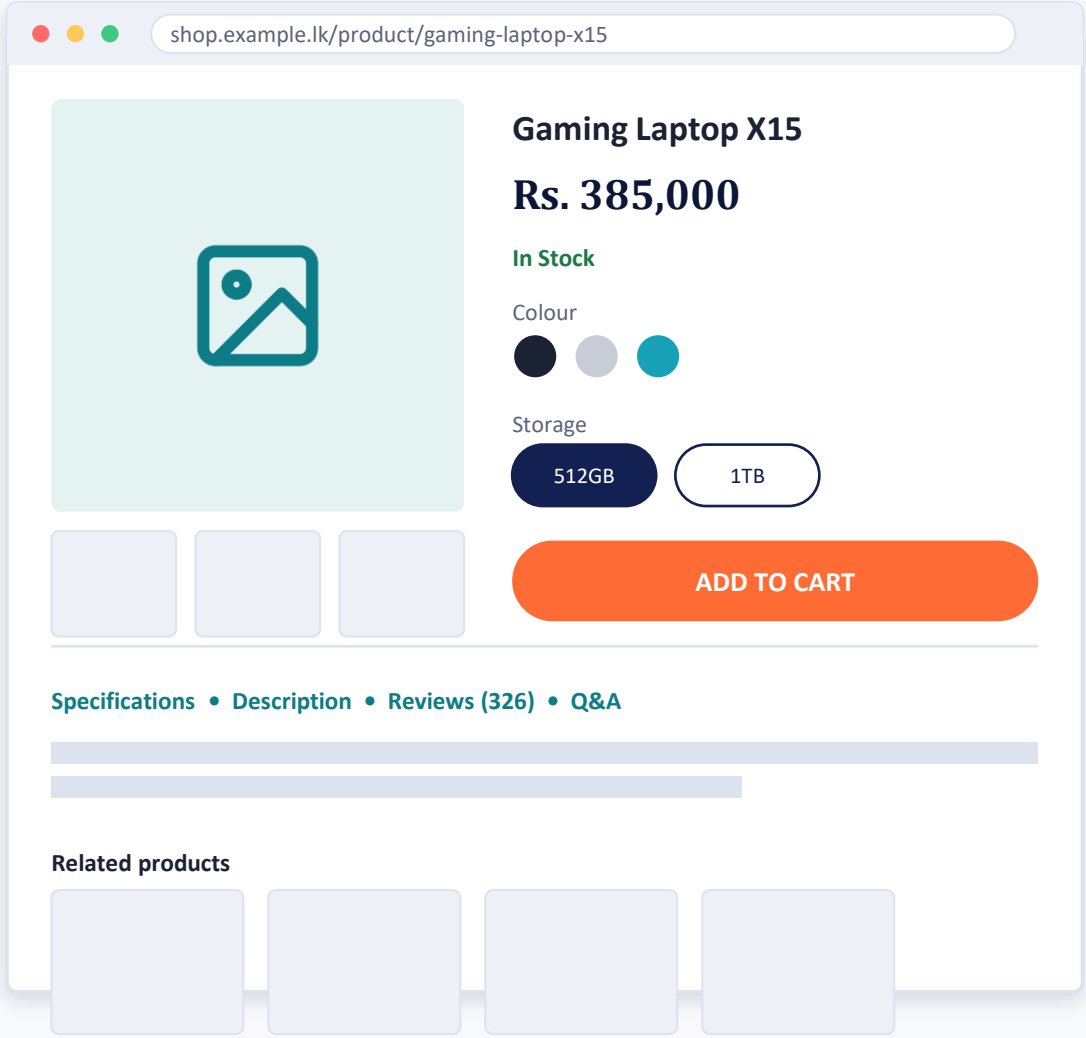
Listing & Detail Page Design

PRODUCT LISTING PAGE (PLP)

- Grid or list view of products within a category
- Product image, name, price and rating snapshot
- Quick-add-to-cart without opening the full page
- Pagination or infinite scroll for large result sets

PRODUCT DETAIL PAGE (PDP)

- Image gallery with zoom, plus variant selectors
- Title, price, stock status and a clear primary CTA
- Description, specifications, and customer reviews



Search Features — Making Products Findable

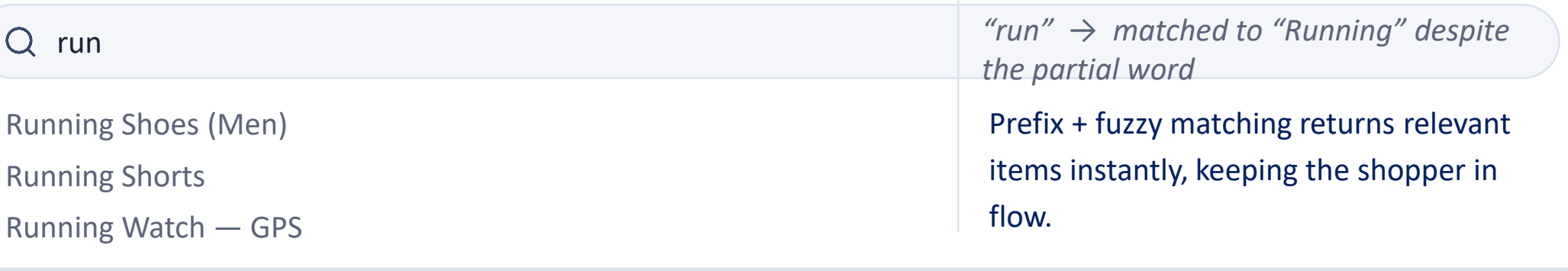
Autocomplete
Suggests matching products and categories as the user types, before pressing enter.

Typo Tolerance
A search for “sneekers” still returns results for “sneakers”.

Relevance Ranking
Ordered by relevance, popularity and current stock availability.

Visual & Voice Search
Emerging ways to search using a photo or a spoken query.

No-Results Handling
Suggest close alternatives instead of a dead end.



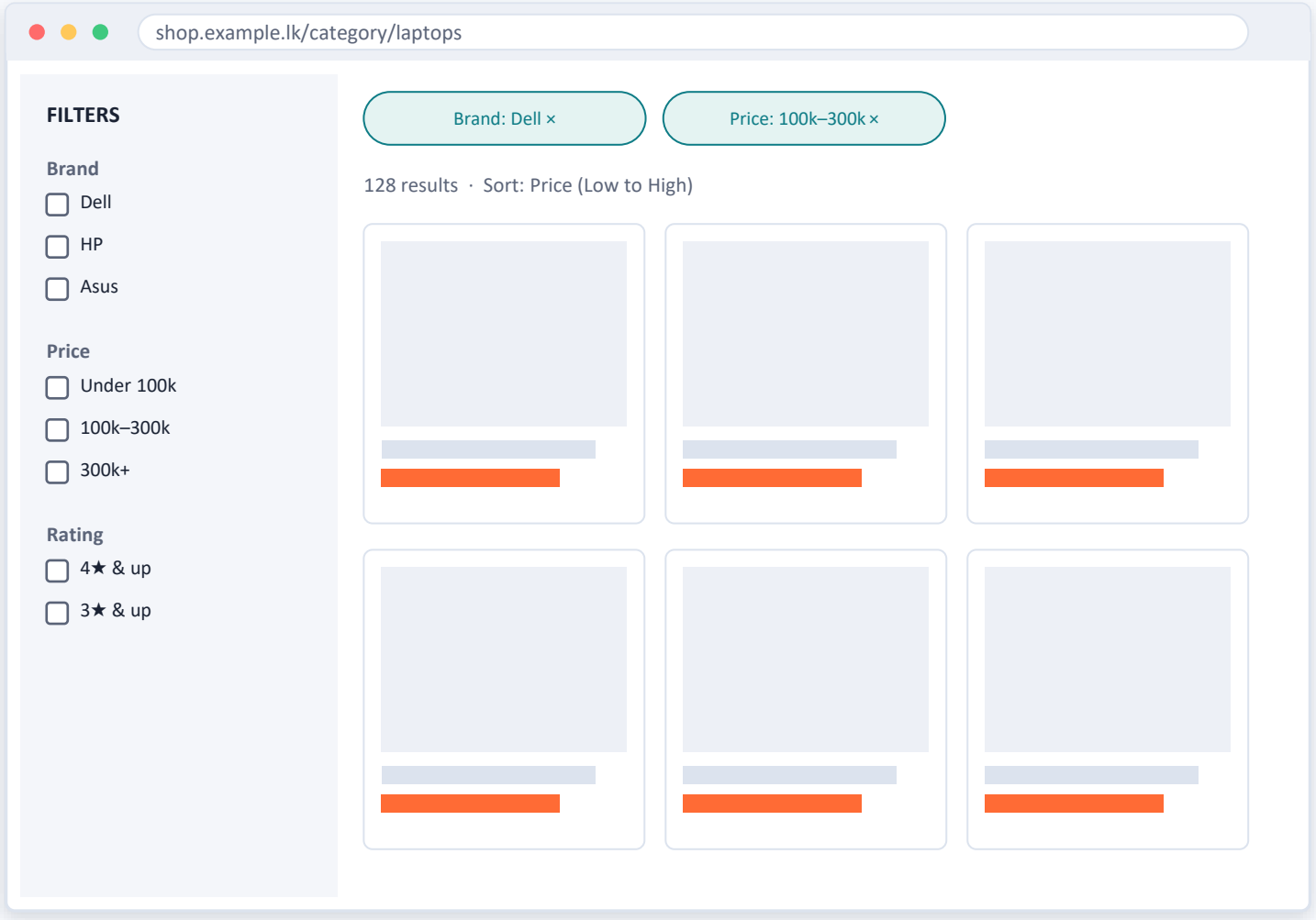
Filtering & Sorting — Faceted Navigation

Facets
Price range, brand, size, colour, rating and availability.

Sorting
Relevance, price (low–high), newest, best-selling.

Applied Filter Chips
Active filters shown as removable tags above the results.

Live Result Count
The number of matching products updates instantly.





Responsive & Mobile-First Design

Core Concepts

Fluid Grid Layouts

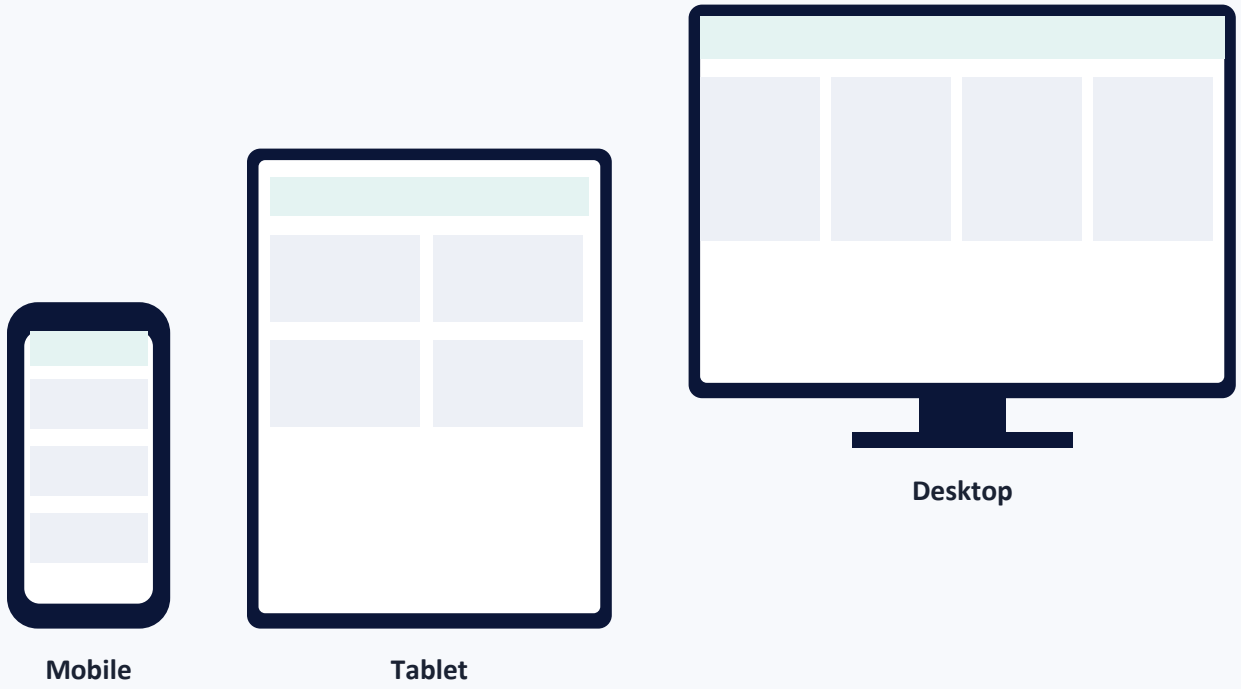
Proportional widths (%, fr) instead of fixed pixel columns.

Flexible Images & Media

Images and video scale to fit within their containers.

CSS Media Queries

Breakpoints adapt the layout to the available screen width.



The Mobile-First Approach

Design for the smallest screen and the biggest constraints first — limited space, touch input, variable network speed — then progressively enhance the layout for larger screens.



Thumb-Friendly Tap Targets

Buttons and links sized at least ~44×44px so they're easy to tap accurately.



Simplified Navigation

Condensed, single-column layouts with a hamburger menu for secondary items.



Streamlined Checkout

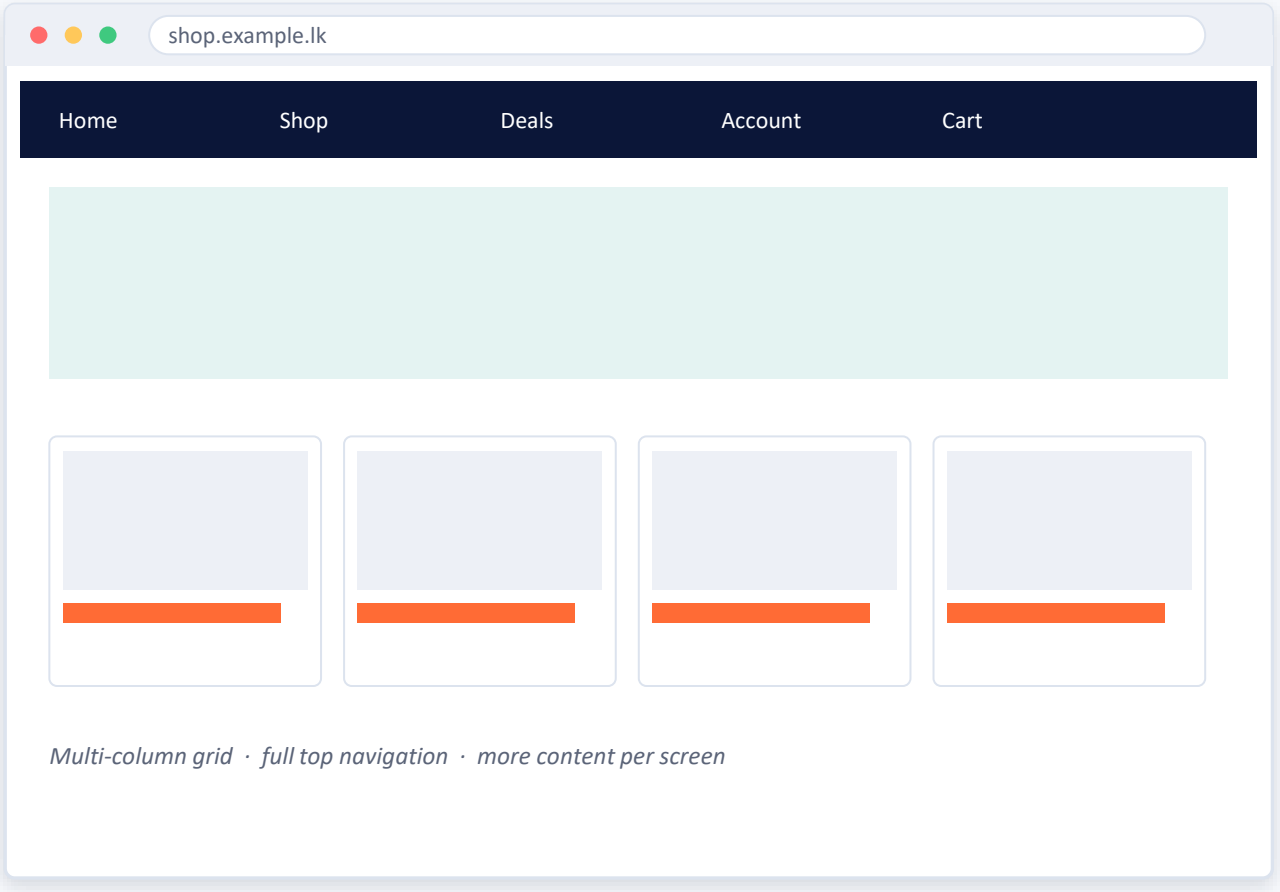
Fewer form fields, autofill, and mobile wallets like Apple Pay or Google Pay.

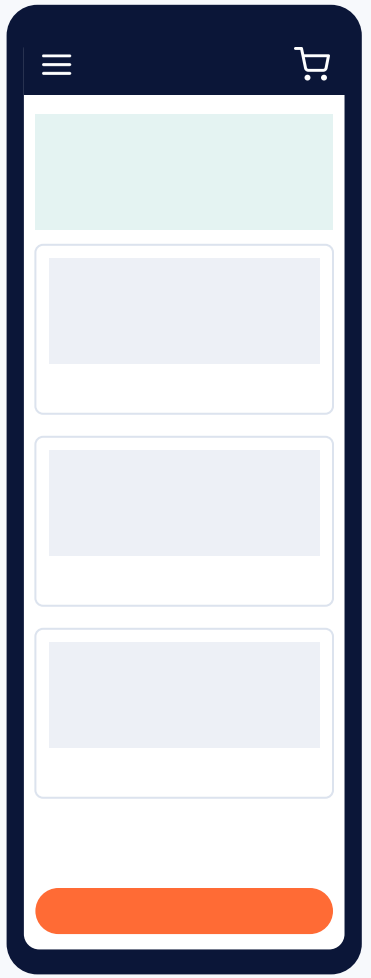


Fast Load Times

Compressed images and lazy loading keep pages quick on mobile networks.

Comparison — Desktop vs. Mobile Layout





KEY DIFFERENCES

- Columns collapse to one
- Menu becomes a hamburger
- CTA sticks to the bottom
- Bigger tap targets, less on screen

Summary

Core Functions	Front-end browsing/checkout and back-end inventory/payments must work as one system.
UX/UI Principles	Simplicity, consistency and visible trust signals turn visits into completed purchases.
Information Architecture	A shallow, logical structure and clear navigation keep products findable.
Catalog, Search & Filters	Well-designed listings, forgiving search and faceted filters help users decide fast.
Responsive & Mobile-First	Design from the smallest screen up — most shoppers arrive on a phone.



Thank You